From Entropy to Epiplexity

Rethinking Information for Computationally Bounded Intelligence

Marc Finzi, Shikai Qiu, Yiding Jiang, Pavel Izmailov, J. Zico Kolter, Andrew Gordon Wilson

Carnegie Mellon University, New York University

arXiv:2601.03220, version 2, 16 March 2026

Speaker: Alex Kravatsky

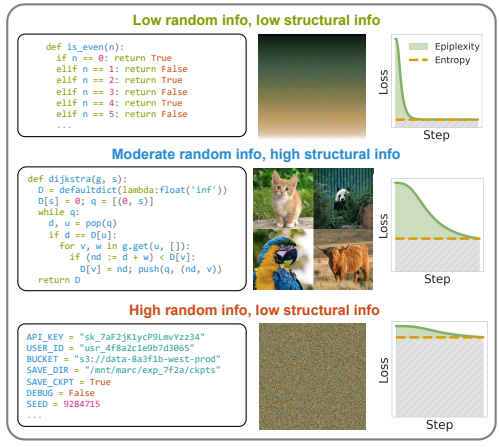
Motivation: evaluating pretraining data

Pretraining is intended to improve performance on tasks beyond the training distribution.

Dataset size and final predictive loss do not quantify how much reusable structure a model acquires.

The paper's question

How can we measure structural information in data relative to a fixed computational budget?



Schematic loss curves. Dashed line: final loss. Green area: cumulative excess loss above that baseline, used as a structural-information proxy. Fig. 1, Finzi et al. (2026).

Shannon entropy and Kolmogorov complexity

Shannon entropy: a distribution

$$p(x) = \Pr(X = x),$$
$$H(X) = \mathbb{E}[-\log_2 p(X)].$$

For discrete X, entropy is a lower bound on the expected length of any binary prefix code.

Uniform n-bit strings: $H(X) = n$.

A fixed string: $H(X) = 0$.

Kolmogorov complexity: a string

$$K(x) = \min_{p:\, \mathcal{U}(p)=x} |p|.$$

The shortest self-delimiting program that outputs x and halts on a fixed prefix-free universal Turing machine $\mathcal{U}$.

An alternating string is specified by its rule and length. Most uniform n-bit strings require at least $n - O(1)$ program bits.

$K(x)$ is uncomputable in general. Definition 1 and Section 2.1, Finzi et al. (2026).

Why introduce a computation bound?

A short program can require a long computation

A simulator and its initial conditions can specify a complex trajectory. Reproducing it may still require more computation than the observer can perform.

Neither Shannon entropy nor unbounded Kolmogorov complexity restricts the runtime of a predictive model.

The paper therefore restricts the admissible models to programs that evaluate probabilities and generate samples within time T.

Within this class, it separates the description of the model from the description of data given the model.

Time-bounded probabilistic models

Fix a prefix-free universal Turing machine $\mathcal{U}$ and a time bound $T(n)$. A program $\mathrm{P} \in \mathcal{P}_T$ implements a distribution P on $\{0,1\}^n$ with two modes:

- **Evaluation:** given x, return $P(x)$ within $T(n)$ steps.
- **Sampling:** given independent random bits, generate $x \sim P$ within $T(n)$ steps.

The returned probabilities must sum to one and agree with the sampling distribution. Roman P denotes the program; italic P denotes its distribution. $|\mathrm{P}|$ is the program length in bits.

Definition 7, Finzi et al. (2026). The formal bound is non-decreasing and time-constructible.

Epiplexity: the two-part description

For X on $\{0,1\}^n$, choose an admissible model minimizing the expected two-part description length:

$$\mathrm{P}^\star \in \arg\min_{\mathrm{P}\in\mathcal{P}_T} \big\{ |\mathrm{P}| + \mathbb{E}[-\log_2 P(X)] \big\}.$$

Among minimizers, select one with the shortest program.

Epiplexity: model description

$$S_T(X) := |\mathrm{P}^\star|.$$

Length of the selected model's program.

Time-bounded entropy: data description

$$H_T(X) := \mathbb{E}[-\log_2 P^\star(X)].$$

Expected data-code length using that model.

$$\mathrm{MDL}_T(X) = S_T(X) + H_T(X).$$

Definition 8, Finzi et al. (2026).

Examples: random bits and a repeated pattern

Assume the length n is specified and the time bound permits the models below.

Uniform bits: $X \sim U_n$

A constant-length program assigns $P(x) = 2^{-n}$ to every n-bit string.

$$\mathrm{MDL}_T(X) \le n + c, \qquad H_T(X) \ge H(X) = n.$$

Consequently $S_T(X) \le c$: a large data-code length can accompany a small model description.

Alternating strings: two equally likely outcomes

$X = 0101\ldots$ or $1010\ldots$, each with probability $\frac{1}{2}$. A constant-length program generates either pattern in linear time.

$$S_T(X) = O(1), \qquad H_T(X) = O(1).$$

Here both the model description and the residual data code are small.

Cryptographic consequences of the time bound

- **Pseudorandom generator.** G maps k random seed bits to $n > k$ output bits. $H(G(U_k)) \leq k$, yet for a secure generator $H_{\mathrm{Poly}}(G(U_k)) \approx n$ and $S_{\mathrm{Poly}}(G(U_k)) = O(1)$.
- **One-way function.** Computing $f(x)$ is efficient; finding a preimage of $f(x)$ is hard for efficient algorithms. For a one-way *permutation*, the paper proves that the two factorization orders have different time-bounded entropies.

Theorems 9 and 13, Finzi et al. (2026). These results assume security against non-uniform probabilistic polynomial-time algorithms.

Conditional epiplexity: predicting labels from inputs

For supervised data (X, Y), the decoder already has the input X. The model evaluates $P(y \mid x)$ and samples $Y \mid X = x$ within the bound for each fixed x.

$$\mathrm{P}^{\star}_{Y|X} \in \operatorname*{arg\,min}_{\mathrm{P} \in \mathcal{P}^{X}_{T}} \{|\mathrm{P}| + \mathbb{E}_{X,Y}[-\log_2 P(Y \mid X)]\}\,.$$

$$S_T(Y \mid X) := |\mathrm{P}^{\star}_{Y|X}|, \qquad H_T(Y \mid X) := \mathbb{E}_{X,Y}[-\log_2 P^{\star}(Y \mid X)].$$

Example: image classification

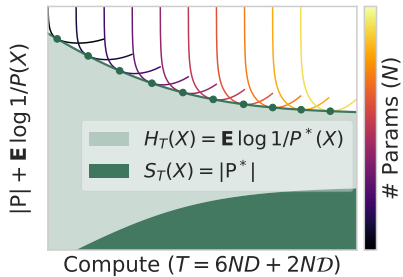
The model code describes the classifier. The data code describes the labels given the images; it does not encode the images.

These quantities are defined directly. In general, $H_T(Y \mid X) \neq H_T(X, Y) - H_T(X)$.

Definition 11, Finzi et al. (2026).

From programs to neural-network estimates

- Search over all programs is intractable, so fix an architecture and optimiser.
- Approximate dense-model accounting: $6ND + 2N\mathcal{D} \leq T$, where N counts parameters and $D, \mathcal{D}$ count training and evaluation tokens.
- Each run gives a model-part estimate plus a validation-data code. Select the shortest admissible two-part code; a lower hull smooths the empirical curve.
- Report its model and data parts as estimates $\widehat{S}_T$ and $\widehat{H}_T$.

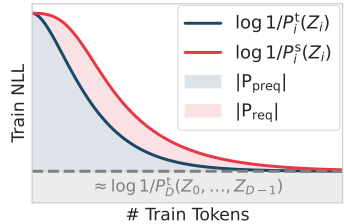


Schematic. Thin curves: training runs, coloured by N. Dots: the lower envelope. Dark green: S_T; light green: H_T. Source: Fig. 2b of Finzi et al. (2026).

Estimating the model part: prequential coding

- P_i has seen $Z_0, \ldots, Z_{i-1}$; P_0 is the initial model.
- Encode Z_i using its predicted probability, then update. The decoder repeats the update.
- The sequential data code sums these costs. Subtract the final-model data code to estimate the model part:

$$\widehat{K}_{\text{preq}} = \sum_{i<M} \left[-\log_2 P_i(Z_i) + \log_2 P_M(Z_i) \right].$$



Blue area: prequential model-part estimate. Red area: teacher–student loss gap used to visualise requential coding. Fig. 2a, Finzi et al. (2026).

- On an i.i.d. stream, the pre-update loss estimates loss on a fresh sample.
- The area is a heuristic model-part estimate; subtraction alone gives no model-only code.

An explicit model code: requential coding

- A teacher model trains on real data; a student model trains on teacher-distributed samples (Qiu et al., 2026).
- Both sides share the student and a random seed. Send the index of a student proposal selected to have the teacher's distribution.
- Relative entropy coding bounds the expected message length by a function of the teacher–student KL divergence:

$$\mathbb{E}[\ell_i \mid \mathcal{F}_i] \le \mathrm{KL}_i + 2\log_2(1 + \mathrm{KL}_i) + \kappa, \quad \kappa < 5.21,$$

Here $\mathrm{KL}_i = \mathbb{E}_{Z\sim P_i^{\mathrm{t}}}[\log_2(P_i^{\mathrm{t}}(Z)/P_i^{\mathrm{s}}(Z))]$ is the Kullback–Leibler divergence, and $\mathcal{F}_i$ is the prior history. $\sum_i \mathrm{KL}_i$ is the leading model-code estimate.

	Prequential	Requential
Model part	heuristic subtraction	explicit student reconstruction
Training	one network	teacher and student, 2–10× slower
Reported values	larger in the paper	smaller in the paper

The bound shown is from Qiu et al. (2026), Appendix A.1; the epiplexity paper used an earlier overhead expression.

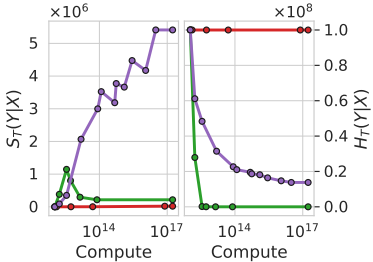
Cellular automata: simple dynamics, different model parts

An *elementary cellular automaton* (ECA) updates binary cells in parallel from their three-cell neighbourhoods (Wolfram, 2002).

Setting. 64 periodic cells; $Y = F_r^{48}(X)$; transformer sweep; requential code.



Rule	15	30	54
$S_T(Y \mid X)$, bits	2×10^5	≈ 0	5.4×10^6
$H_T(Y \mid X)$, bits	≈ 0	10^8	1.5×10^7

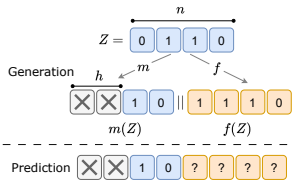


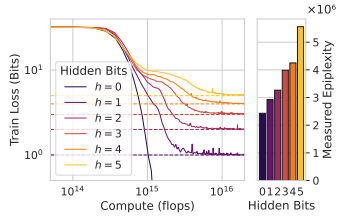
Left: rollouts, time downward. Right: S_T and H_T against compute in FLOPs. Table: our estimates from the plots at 10^{17} FLOPs. Source: Fig. 3 of Finzi et al. (2026).

Under this protocol, rule 54 has the largest estimated model part.

Hidden variables: generation and prediction differ

Setting. Hide h bits of a random $Z \in \{0,1\}^{32}$; predict four rule 30 updates of Z from the visible bits.





Left: the data. Right: training loss against compute in FLOPs (dashed: h bits) and S_T for each h. Source: Fig. 5 of Finzi et al. (2026).

- Generation uses the complete state Z. Prediction conditions on only its visible bits.
- As h increases, reaching a loss near h bits requires more compute, and the estimated model part increases.

Prediction requires a distribution over hidden states; generation does not.

Experiment: the order of chess data

Setting. Lichess games in two orders: moves → final board, or board → moves. Transformer sweep with requential coding.



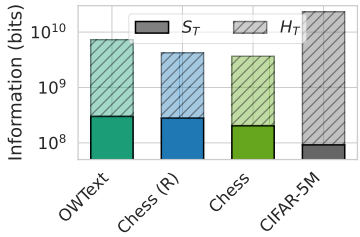
Left: H_T and S_T against compute in FLOPs. Right: S_T and accuracy after fine-tuning. Source: Figs. 4c and 7 of Finzi et al. (2026).

- At large compute, reverse order gives higher H_T and S_T.
- After fine-tuning, reverse-order models reach 0.28 board-evaluation accuracy versus 0.21 for forward-order models; puzzle accuracy is similar.

This supports the authors' hypothesis that reverse ordering develops more useful board-state representations.

Application: structure in natural data

Setting. Equal token budgets and a shared compute bound. Compare OpenWebText, chess and CIFAR-5M with the requential code.



Chess (R): reverse order. Source: Fig. 8a of Finzi et al. (2026). Table: our estimates from the figure.

Dataset	S_T, bits	MDL_T, bits	S_T/MDL_T
OpenWebText	3×10^8	7×10^9	4%
Chess, reverse	2.8×10^8	4×10^9	7%
Chess, forward	2×10^8	3.5×10^9	6%
CIFAR-5M	9×10^7	2.3×10^{10}	0.4%

- Under this protocol, text has the largest estimated model part; pixel images have the largest residual part.
- These measurements compare representations at a fixed token count, rather than equal bytes or semantic content.

This ranking is relative to the tokenisation, model family and compute budget.

Conclusions and limitations

Summary

- S_T counts selected model bits; H_T is the expected data-code length.
- Computational restrictions can change both the total description length and its decomposition.
- Neural-network estimates combine a model-code estimate with validation loss, optimized under a compute budget.

Limitations

- S_T is not a guarantee of downstream usefulness.
- The existence theorem is conditional on cryptographic assumptions and proves $\Omega(\log_2 n)$ growth.
- Estimates depend on architecture, optimiser, T, dataset size and random seed.

Literature

M. Finzi et al. *From Entropy to Epiplexity*. arXiv:2601.03220v2, 2026.

S. Qiu et al. *Requential Coding*. arXiv:2607.11883, 2026.

M. Li, P. Vitányi. *Kolmogorov Complexity*. Springer, 2008.

S. Wolfram. *A New Kind of Science*. 2002.

Y. Jiang et al. Adaptive data optimization. *ICLR*, 2025.

Questions for the audience: answers

1. Let $X \sim U_n$. Which quantity is close to n: $S_T(X)$ or $H_T(X)$?

Answer

$H_T(X) = n + O(1)$. A constant-length uniform model gives $\mathrm{MDL}_T(X) \leq n + c$; since $H_T(X) \geq n$, the selected model has $S_T(X) \leq c$.

2. In the prequential estimate, which part of the training-loss curve approximates the epiplexity?

Answer

The cumulative excess loss above the final-model baseline: $\sum_i [-\log_2 P_i(Z_i) + \log_2 P_M(Z_i)]$.

3. What does changing T change?

Answer

It enlarges the admissible model class. The minimum total MDL_T cannot increase, while the selected model length S_T may rise or fall.

Backup: basic properties

Proposition 1 (Sec. 3 of the paper)

Let c_1, c_2 be constants independent of X. For a random variable X on $\{0,1\}^n$:

1. $S_T(X) \geq 0$ and $H_T(X) \geq 0$;
2. $H(X) \leq \mathrm{MDL}_T(X) \leq n + c_1$;
3. $\mathrm{MDL}_{T'}(X) \leq \mathrm{MDL}_T(X)$ whenever $T'(n) \geq T(n)$;
4. $\mathrm{MDL}_{T'}(f^{-1}(X)) \leq \mathrm{MDL}_T(X) + |\mathrm{f}| + c_2$ with $T'(n) = T(n) + \mathrm{Time(f)}$, for a program f that computes a bijection f in fixed time.

Property 4 is the analogue of $K(f(x)) \leq K(x) + K(f) + c$. Under a time bound a short program for f^{-1} gives no short program for f.

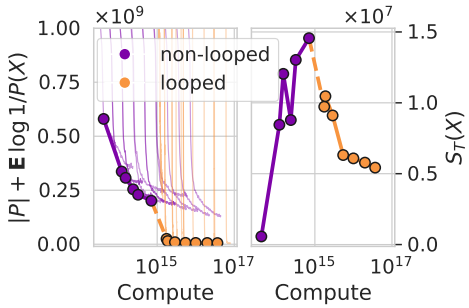
Theorem 1 (Thm. 10 of the paper)

If one-way functions secure against probabilistic polynomial-time algorithms with polynomial advice exist, then some random variables X_n on $\{0,1\}^n$ have $S_{\mathrm{Poly}}(X_n) = \Omega(\log_2 n)$.

Backup: epiplexity need not increase with compute

Setting. ECA rule 54, 64 updates. The non-looped model predicts the final state; the looped model simulates each update.

- Below 10^{15} FLOPs, the non-looped model gives the shorter code; its model part reaches 1.45×10^7 bits.
- Above that bound, the looped model wins because the rule is a compact model.
- Thus epiplexity need not increase with the budget; growth is expected to be more typical for natural data.



Two-part code and S_T against compute in FLOPs. Source: Fig. 6 of Finzi et al. (2026).

With less compute the best model is a longer program: S_T need not grow with T.